Quiz 13 Solutions

Give exact answers and justify your reasoning. Do not use electronic tools. All functions are real-valued, angles are measured in radians, and all integrals are Riemann integrals.

1. Find the domain and derivative of

$$H(x) = \int_0^{\ln x} \sqrt{1+t^4}\, dt$$

without evaluating the integral.

Solution. The real logarithm requires $x > 0$. The integrand is continuous for every real t, so no further restriction is needed; oriented integrals also allow $\ln x < 0$. Set $B(u) = \int_0^u \sqrt{1+t^4}\, dt$. FTC I gives $B'(u) = \sqrt{1+u^4}$, and $H(x) = B(\ln x)$. Hence

$$\boxed{\operatorname{dom} H = (0,\infty), \qquad H'(x) = \frac{\sqrt{1+(\ln x)^4}}{x}.}$$

2. Let f be continuous on $[0,1]$, and suppose $\int_0^1 f(x)\,dx = 2$. Prove that

$$\int_0^1 f(x)^2\,dx \geq 4,$$

and give a function for which equality holds. You may use $(f(x)-2)^2 \geq 0$.

Solution. The square is continuous and nonnegative. Comparison and linearity give

$$\begin{aligned} 0 &\leq \int_0^1 (f(x)-2)^2\,dx \\ &= \int_0^1 f(x)^2\,dx - 4\int_0^1 f(x)\,dx + \int_0^1 4\,dx \\ &= \int_0^1 f(x)^2\,dx - 8 + 4. \end{aligned}$$

Therefore $\int_0^1 f(x)^2\,dx \geq 4$. Equality holds for the continuous constant function $f(x) = 2$, whose integral on $[0,1]$ is 2 and whose square has integral 4.